Enhanced Urban Mobility through Deep Learning Based Real-Time Passenger Head Counting and Occupancy Estimation

Project Type: Research-Based Project

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Section: A

1 Project Title

Enhanced Urban Mobility through Deep Learning Based Real-Time Passenger Head Counting and Occupancy Estimation

2 Background of the Project

As the world moves towards smart cities and Intelligent Transport Systems (ITS), Automated Passenger Counting (APC) has become more than a data collection exercise and essential for smooth transport operations [1]. Existing approaches to measuring bus patronage (such as manual counting or ticket-based systems) are time consuming, prone to error and do not account for fare evasion or real-time variations. Real time data on passenger occupancy is crucial for transport operators to optimise fleet allocation, scheduling and passenger management [2]. A lack of knowledge on occupancy leads to health and safety risks on overcrowded buses, and wasted cost (both operational and environmental) on under-serviced ones [3].

Current APC technologies include infrared beams, pressure sensitive floor pads, Wi-Fi probe, and video analytics [4]. In particular, camera based video analytics has gained most interest due to the low cost, ubiquity and versatility of cameras as a sensor option [5]. But the most common approach to camera based APC is line of interest (entry/exit) counting, where individual passengers are counted as they pass through a virtual line at the bus door [6]. These approaches work well in laboratory environments, but are prone to counting error over time, are sensitive to camera angle and occlusion at the bus door, and do not provide a direct measure of occupancy within the bus [7, 8].

Head counting, in which the number of passengers in a scene is estimated directly from a single image or video frame, is an alternative approach. This approach yields direct measures of current occupancy from mounted cameras, rather than inferring it from cumulative entry and exit counts. [9]. This eliminates the issue of counting error accumulation, and enables occupancy to be adjusted along the journey. However, head counting inside buses has been relatively overlooked in research and has never been directly compared to entry/exit based methods in real practice. In addition, current camera based systems can be considered a privacy concern, as conventional cameras (RGB) record biometric data about passengers (this is becoming more important with privacy regulations such as GDPR) [10]. This study aims to fill both of these gaps, by designing a privacy preserving deep learning head counting system for in bus overhead video.

3 Objectives

Main Objective: To create a real time, privacy aware deep learning head counting system for counting people to estimate passenger occupancy in public transport.

Specific Objectives:

1. Design a two-pronged approach using object detection (YOLOv8) for seated areas and density estimation (MobileNetV2) for standing areas.

2. Develop a privacy-focused pipeline considering source desensitization or low resolution depth information to avoid person recognition.

3. Perform field experiments in crowded urban bus environments to evaluate the proposed method and compare it with an entry/exit-based approach.

4 Justification of the project

This project is worthwhile because head counting is a real problem with transportation networks [11]. Counting enables service operators to better control fleet size, schedule so that service frequency is optimal, and enhance transportation quality in general [12]. A head-counting method is promising because it can be less invasive than identifying bodies or faces, but still beneficial for estimating occupancy. Current research shows that bus APC is still challenging with occlusions, varying camera angles and illumination, and that the detection of small heads is still problematic for many systems [13].

5 Expected Outcome

- A deep learning model to count heads from in-bus, overhead video, with measured accuracy metrics (MAE, RMSE, per-scene accuracy) in different conditions.
- An edge deployable prototype system demonstrating real-time feasibility on hardware suitable for bus installation (e.g., Raspberry Pi).

6 Beneficiaries

- Transport operators and bus fleet operators need reliable, real-time passenger load data for fleet planning, route planning and service planning.
- Public transport authorities need passenger occupancy data for planning infrastructure and services.
- Passengers, who indirectly benefit from improved transport services and, directly, from increased privacy when travelling on public transport.

7 Contribution

- First empirical study directly comparing the head counting method with line-of-interest (entry/exit) APC in real bus deployment scenarios.
- The design of a privacy-preserving head counting system with anonymity at the feature extraction level (not post processing).
- Rigorous analysis of head counting accuracy loss under varying crowd densities (sparse, moderate, crowded) in the bus environment.

8 Tools

- **Programming Language:** Python
- **Libraries:** OpenCV, TensorFlow
- **Models:** YOLOv8, MobileNetV2

9 Project Schedule

Phase	Duration	Start	End
Literature Review	10 days	April 15	April 27
Dataset Collection & Annotation	15 days	April 25	May 09
Baseline Model Development & Training	18 days	May 10	May 27
Counting Logic and Tracking Integration	10 days	May 28	June 06
Testing and Validation	12 days	June 07	June 18
Edge Deployment & Optimization	12 days	June 19	June 30

Section: B

1 Research Motivation

The existing APC systems have the ”Line of Interest” problem where missed detections at the bus entry and exit points cause persistent errors, resulting in compounded errors at multiple stops [14, 15]. Also, traditional computer vision approaches fail to track under ”crowd-intensive” bus conditions due to heavy occlusion, where only heads are visible [16]. We urgently need a system that can interpret ”crowd density” rather than just individual motions, particularly in the new privacy regulations as in GDPR.

2 Brief review of works related to the proposal

Automatic Passenger Counting systems are essential for public transport systems as they help track passenger movement, increase efficiency and inform decision-making. These technologies are extensively deployed in buses, trains and metro services for real-time occupancy estimation. Historically, passenger counts have been recorded manually or from ticketing systems, which can be inaccurate, labour-intensive and not real-time. As a consequence, there has been a growing interest in research on automatic passenger counting systems.

To overcome these challenges, some sensor-based and vision-based methods have been explored. Sensor-based counting systems, such as infrared sensors, radio-frequency identification and pressure-sensitive mats, are typically employed to monitor passenger movements. Although these methods are straightforward, they have a number of limitations, such as high installation and maintenance expenses, susceptibility to environmental factors like bus engine vibrations causing data blurring, and decreased accuracy in dense environments [17, 18,]. Consequently, vision-based approaches have gained popularity.

Vision-based passenger counting systems rely on cameras to track and count passengers. These techniques are used to detect and track people and count them as they board or alight. Vision-based systems are more flexible and informative than sensor-based systems but suffer from problems like occlusion, lighting, and densely populated environments which can impact counting accuracy [19, 4,]. As artificial intelligence technology advances, vision-based systems that have incorporated deep learning and hybrid methods have become popular. The use of deep learning models, such as convolutional neural networks and real-time object detection algorithms such as YOLO, enhance the detection of passengers [17, 20,]. Moreover, the integration of other sensor information with vision-based approaches improves accuracy. However, these systems can be computationally intensive and may not provide real-time performance on limited computing resources.

Person detection-based techniques try to detect the whole body for counting, while face detection techniques detect facial features. Another popular method is density estimation, which estimates the presence of people based on pixel density. While these approaches have achieved good performance, they can be less accurate in densely populated scenes as a result of occlusion and overlapping people and viewpoints [20]. To address these issues, head counting methods have gained popularity as a better alternative, especially in crowded environments. The use of head detection involves the detection of the upper part of the body, which is less prone to occlusion compared to other body parts. This makes head counting more accurate for passenger counting in crowded public transport vehicles. Several methods, such as deep learning detectors and density-based methods, have been proposed to enhance head detection [7, 4,].

Table 1: Summary of Related Work

Title	Focus	Primary Limitations
Video-based automatic people counting for public transport [4]	Comparing on-bus vs. off-bus deployment using MobileNetV3 SSD.	Susceptible to weather (rain), direct sunlight, and occlusion from umbrellas.
A Methodology for Empirically Evaluating Passenger Counting Technologies [18]	Comparison of Video, WiFi, 3D IR, and Pressure Mats.	IR sensors struggled with engine vibration; mats had mechanical installation issues at front doors.
NAPC: A Neural Algorithm for Automated Passenger Counting [10]	3D LiDAR recordings with LSTM RNN for privacy-friendly counting.	High labor cost for manual annotation; dataset specifically excludes children.
Estimation of the Number of Passengers in a Bus Using Deep Learning [20]	Hybrid YOLOv3 and CAE for density estimation in crowded buses.	Blind spots in corner seats where only lower bodies are visible to rear cameras.
Passenger Detection and Counting for Public Transport Systems [19]	GMM, SVM, and Kalman Filter for line-of-interest counting.	Prone to false detections from shadows, sudden motion, and lighting changes.

In summary, the literature demonstrates that although video-counting solutions are adaptable and scalable, they are dependent on conditions, camera angles and density [1]. Current head counting approaches still need to overcome issues of computational efficiency, varying lighting conditions, and practical deployment. Thus, the aim of this study is to overcome these challenges by proposing a reliable head-based passenger counting system, which will enhance the accuracy, scalability and practicality of the system.

3 Research Question

- Can head detection be used to count passengers accurately in bus interiors?
- How does camera position affect head-counting accuracy in crowded buses?
- Which detection model performs best for small-head detection in real bus scenes?
- Can tracking improve counting stability across frames and reduce double counting?
- How robust is the system under low light, occlusion, and motion blur?

4 Methodology

4.1 Input Video

The first step is to record a live video feed from onboard cameras in public transport vehicles. The cameras are usually located on the roof or near the doors to capture the heads of the passengers.

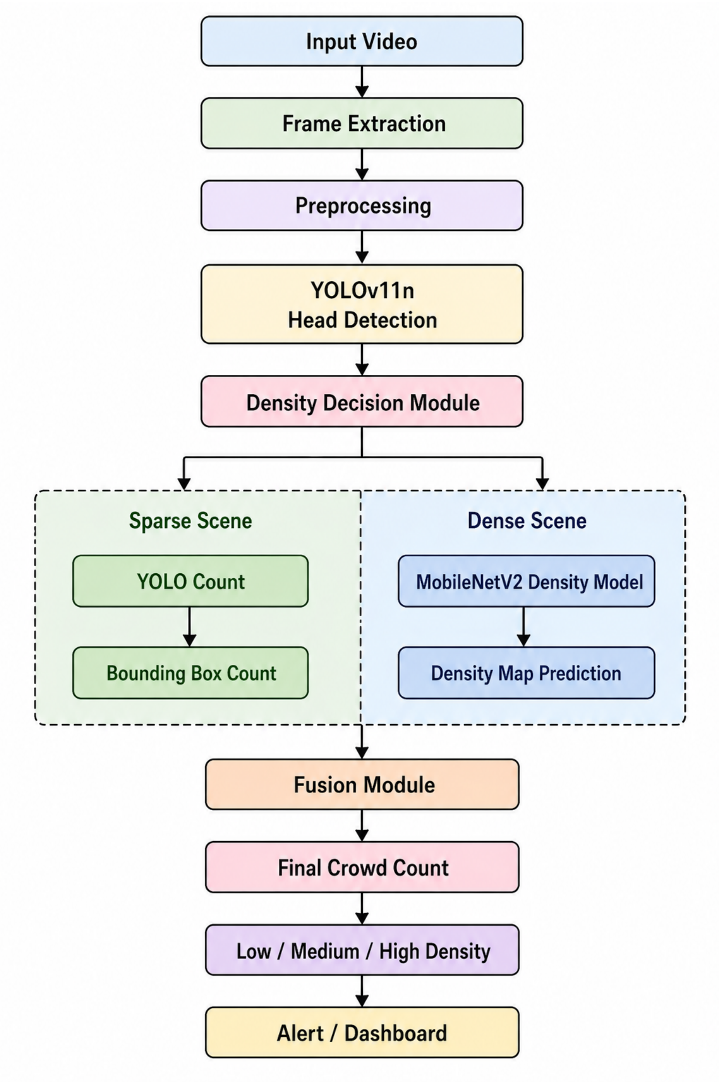

Figure 1: System Architecture

4.2 Frame Extraction

After the video stream is established, it is broken down into frames that can be processed individually. Videos are simply a series of images, and converting them to frames allows the system to process frames at discrete moments in time.

4.3 Preprocessing

The frames are preprocessed to improve their quality and prepare them for deep learning models. This can involve resizing, normalizing pixel intensities, and applying noise reduction or contrast amplification.

4.4 Head Detection using YOLO

These frames are then fed to an object detection model like YOLOv8. These models are capable of real time object detection and can detect either people or heads.

4.5 Density Decision Module

Once the people are detected, the scene is assessed for its crowd density via a decision module. This module takes into account the number of heads detected, their arrangement in space, and the overlap of the bounding boxes. Using these features, the system assesses whether the scene is sparse (small number of passengers and clear separation) or dense (large number of passengers and overlap/occlusion).

4.6 Sparse Scene (YOLO Counting)

If the scene is sparse, the system uses the detections directly to count the number of passengers. Each person will be represented by a bounding box and the number of passengers will be equal to the number of detected heads.

4.7 Dense Scene Estimation (MobileNetV2)

If the scene is determined to be dense, the system adopts a density estimation approach with a convolutional neural network (CNN) model, such as MobileNetV2. Rather than localising heads, this network produces a density map, in which the intensity of each pixel indicates the density of people in that area of the scene. The number of passengers can be estimated by integrating the density map.

4.8 Fusion Module

Finally, the sparse and dense count estimates are fused in a module to provide a more reliable estimate. At its most basic, the system will simply choose between the output of the detection based or density based method based on the scene classification.

4.9 Final Crowd Count

The fusion process outputs a single number that is the total passenger count in the vehicle at the scene.

4.10 Density Classification

The total number of passengers is also classified into qualitative density terms such as low, medium and high. These levels are set by established thresholds, based on vehicle capacity.

4.11 Alert and Dashboard System

The last step is to present the processed data on a monitoring dashboard or provide alerts. The dashboard can display real-time passenger numbers, density, and historical data for comparison. Overcrowding alerts can be triggered to assist transport authorities in making decisions.

5 Dataset Information

5.1 ShanghaiTech Crowd Counting Dataset

The ShanghaiTech Crowd Counting Dataset is one of the most widely used benchmarks for crowd counting, density estimation, and crowd localization in computer vision. It was introduced in the influential paper Single-Image Crowd Counting via Multi-Column Convolutional Neural Network (MCNN) and has become a standard dataset for evaluating crowd-counting algorithms.

6 Risk and Challenges

Risk	Severity	Mitigation Strategy
Occlusion in dense crowds degrading head detection	High	Hybrid detection + density estimation, density adaptive model switching.
Privacy processing removing too much spatial info for counting	Medium	Compare privacy-preserving methods to evaluate the best trade-off between privacy and crowd counting accuracy.
Lighting variation affecting detection accuracy	Medium	Add lighting to dataset; try histogram equalisation and HDR pre-processing.
Camera installation restrictions on commercial buses	Low	Partner with operator; magnetic mount install.

Section: C

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Section: D

Declaration

I do hereby declare that all the information in this proposal is correct and complete. The proposal is completely written by me, not AI-generated.

Md Masum Billah

Approval

I hereby declare that I am fully aware of this project and have full confidence in the originality of this proposal. I am approving this proposal for submission to the Project Committee.

Nazmun Nahar
Assistant Professor